Computer Engineering

Performance Evaluation of Computer Systems and Networks

Evaluation of Highway Barrier performance with ETC lanes

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1. INTRODUCTION

Problem Description

A highway toll barrier has N lanes, K of them are dedicated exclusively to vehicles equipped with an electronic toll collection (ETC) device only. A percentage p of vehicles is equipped with an ETC device and vehicles arriving at the barrier are either cars or trucks, which have their own arrival rate exponentially distributed. The service time of the "Standard lanes" for the trucks are 1.5 times the service time of the cars, while the service time of the "ETC lanes" is constant.

In both cases, vehicles select the lane with the smallest queue among those they are permitted to use.

The percentage p of vehicles equipped with the ETC can be varied and so does K.

We focus on comparing the queueing and response time of the two options under a varying workload.

At least the following scenarios must be evaluated:

- Exponential distribution of service times for standard lanes.
- Lognormal distribution of service times for standard lanes.

Objectives

The main objective of the project is to study the behavior of the whole highway toll barrier with different combinations of K and p, under varying workload.

Performance indexes:

To assess the performance of the system we took as a reference:

- The Mean Response Time E[R]: defined as mean time between the arrival of a vehicle at the barrier and the end of its service.
- The Mean Waiting Time E[W]: defined as mean time between the arrival of a vehicle at the barrier and the start of its service.

2. MODELING

General Assumption

These are the general assumptions that have been made:

- The vehicle of the highway coincides with the job of the system.
- A vehicle is assigned to the lane with the least number of vehicles in the queue and it cannot change its lane.
- The tills are represented as FIFO queues of unlimited capacity.
- In case of a tie between two or more lanes, the lane assigned will be the one with the minimum id.
- The service time of the "ETC lanes" must be higher than zero.
- Except the case where k is zero, the vehicles equipped with the ETC device are assigned exclusively to the "ETC lanes" and those without ETC can only go to the "Standard lanes" (no hybrid lanes).

Preliminary Validation

Before the implementation, preliminary validation is necessary to verify that the model is correct. In order to do so, the previous general assumptions should be analyzed.

- The unlimited capacity of the lanes is a reasonable assumption, because in a real Highway toll barrier vehicles could queue themselves along the whole highway, thus making the queues approximately unlimited.
- In the case of a tie between two or more lanes, the assignation to the lane with the minimum id is a reasonable assumption because humans would choose the first lane unconsciously.
- The assumption that a vehicle equipped with ETC device is assigned only to the ETC lanes (except in the case where k is zero) is a reasonable assumption because in the real world it is the best choice in terms of service time.
- The service time of the "ETC lanes" higher than zero is a reasonable assumption because it is considered the slowdown time to respect the maximum limit of velocity in proximity of the Highway toll barrier.

Factors

The following factors may affect the performance of the system:

- **N**, number of total lanes of the Highway toll barrier.
- **K**, number of the lanes dedicated to vehicles equipped with the ETC device.
- **p**, percentage of vehicles equipped with the ETC device.
- **T_c**, mean interarrival rate for cars.
- **T_t**, mean interarrival rate for trucks.
- **S**, mean service time of “Standard lanes”.
- **S_etc**, service time of “ETC lanes”.
- **stddev**, standard deviation of the lognormal distribution

3. IMPLEMENTATION

Modules

The following modules have been defined:

- **Highway**: simple module which generates cars and trucks, according to the specified distributions, and sends them to the Barrier module as messages.
- **Barrier**: compound module which represents the system under analysis. It is composed by:
    - **Switch**: simple module which routes the received messages in a suitable lane for its type.
    - **Lane**: simple parent module that simulates the behavior of an highway lane by implementing a FCFS policy.
    - **StandardLane**: simple module that extends the parent module Lane.
    - **EtcLane**: simple module that extends the parent module Lane.

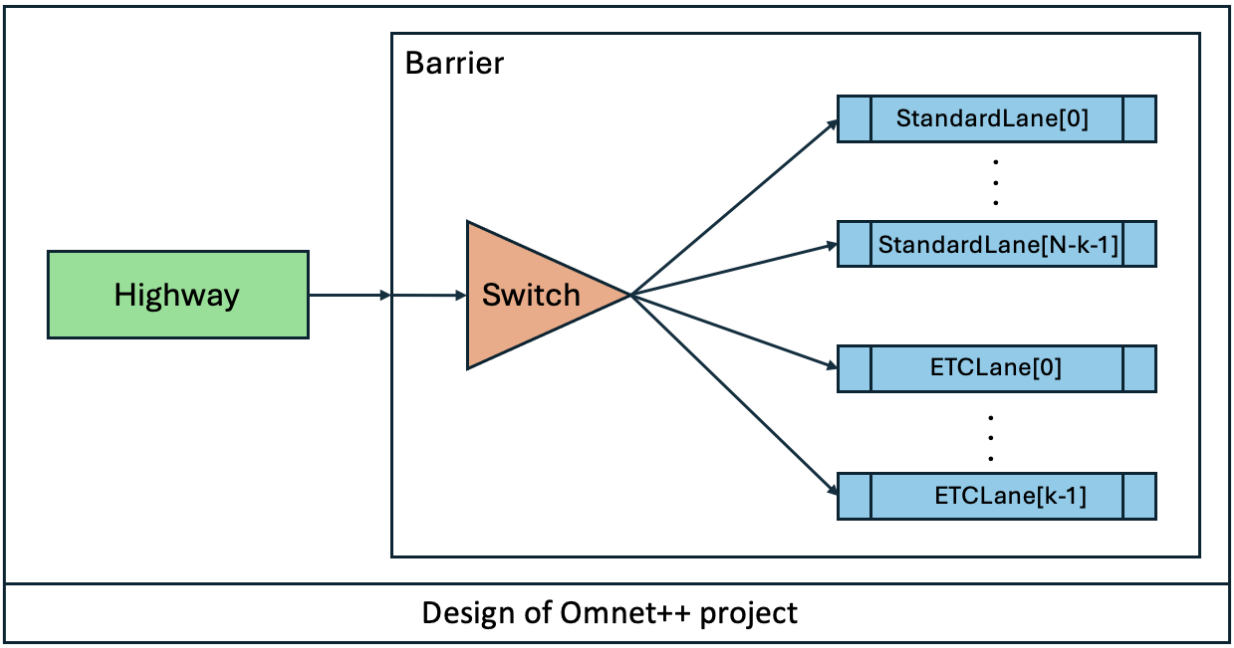


Figure 3.1 - Graphic representation of the implementation of Omnet++

Messages

We have defined a new format of messages to properly represent cars and trucks, called *Vehicle*. Its fields are:

- bool **isCar**: boolean describing if the message represents a car or a truck.
- bool **hasEtc**: boolean describing if the vehicle is equipped with the ETC device.
- simtime_t **arrivalTime**: used to record the arrival time of the vehicle at the barrier.

Modules' behavior

Highway:

- Generates a random number from the interarrival distribution of cars/trucks (exponentially distributed) and sets a timer accordingly.
- When the timer expires, generates a uniformly distributed random number to determine, based on the percentage *p*, if the vehicle is equipped with the ETC device.
- Sends a message to the Barrier module with the fields *isCar* and *hasEtc* properly instantiated.

Switch:

- Receives the message from the Highway module and checks if it is equipped with the ETC device.
- Among the admissible types of lanes based on the presence of the ETC device, selects the lane with the least number of vehicles. If there is a tie, select the lane with minimum id.
- Sends the message to the chosen lane.

Lane:

- Receives a message from the Switch module, records its arrival time in the *arrivalTime* field of the message and pushes it in a FIFO queue. If the queue was empty before this, start processing the message.
- Generates a random number for the vehicle's service time, with the distribution depending on the type of lane and the type of vehicle, and sets a timer accordingly. Emits the waiting time of the vehicle.
- When the timer expires the service concludes, emits the response time, removes the vehicle from the queue and checks if there is another vehicle in the queue.

StdLane / EtcLane:

- Generates a random number for the service time using different distributions for each module.
- Inherit all other functions from the parent module Lane.

Random number generators

Each type of vehicle (car and truck) needs to use two random numbers:

1. The first one has a uniform distribution and, comparing it with the percentage p, it is used to determine if the vehicle is equipped with the ETC device.
2. The second one has an exponential or lognormal distribution and it is used to generate the vehicle arrival rate.

Using two different RNGs for cars and two RNGs for trucks would still be correct, but using too many RNGs could make computations too slow. Let's see if we can do better.

In our code we generate first the random number for p and, right after, the random number for the next arrival of the vehicle. Since the generation of these two numbers is always coupled, we can use only one single RNG for each type of vehicle (changing the distribution of the arrival rate random numbers will not interfere with the generation of the other random number because they always go in sequence). Now we can say that we are using one RNG for cars and one for trucks.

We used another RNG for the service distribution of all the lanes. It means we are using 3 total RNGs (using 5 of them is still correct, but less efficient).

4. VERIFICATION

The following sections will present a series of verification tests designed to evaluate the system's predictable behavior. Each test will specify the modifications made to the general configuration, which consists of the following parameters:

- **N** (Total lanes) = 8
- **K** (ETC lanes) = 2
- **p** (percentage of vehicles equipped with ETC device) = 0.3,
- $\mathbf{T_c}$ (Mean interarrival time of cars) = 9s
- $\mathbf{T_t}$ (Mean interarrival time of trucks) = 90s,
- **S** (Mean service time for standard lanes) = 62s,
- **S_etc** (Service time for ETC lanes) = 5s .

Degeneracy test

An analysis of model behavior under degenerate conditions, achieved by setting parameters to 0 or extreme values, ensuring that the system consistently produces the expected results. We devised the following tests:

if **K = 0**, the Highway barrier has only standard lanes. In this case, vehicles equipped with the ETC device are assigned to standard lanes, so the value of p is irrelevant. The case of K = N was not considered because it's unrealistic and meaningless for our goals. Furthermore, our code does not handle this case.

If **p = 0** and **0 < K < N**, the Highway barrier has both standard and ETC lanes. However, no vehicles will be equipped with ETC devices and thus assigned to the ETC lanes, rendering them useless. If K is large, there's the risk that N-K standard lanes are not sufficient to handle all the incoming vehicles.

Consistency test

The consistency test aims to confirm predictable and consistent behavior between the system and its generated output. We designed two different tests, each focusing on different parameter changes:

Consistency test A

This test consists of two situations in which only one type of vehicle arrives at the barrier: in the first one all the vehicles are cars while in the second one all the vehicles are trucks. To model this case, it is enough to set the time arrival of the type of vehicle excluded to a very large value. Furthermore, we set K = 0 and p = 0 to prevent the use of ETC lanes, where the service time is constant regardless of the vehicle type.

Test A.1 - only cars

- **N** (Total lanes) = 20,
- **K** (ETC lanes) = 0,
- **p** (percentage of vehicles equipped with ETC device) = 0,
- $\mathbf{T_c}$ (arrival time of cars) = 15s,
- $\mathbf{T_t}$ (arrival time of trucks) = 10000s .

Test A.2 - only trucks

- **N** (Total lanes) = 20,
- **K** (ETC lanes) = 0,
- **p** (percentage of vehicles equipped with ETC device) = 0,
- $\mathbf{T_c}$ (arrival rate of cars) = 10000s,
- $\mathbf{T_t}$ (arrival rate of trucks) = 15s .

With the choice of 20 total lanes and a relatively low vehicle interarrival time we aim to avoid queueing in order to assess the correctness of the service times. Given that in standard lanes the service time for trucks is 1.5 times the service time for cars, we expect the same consistent ratio in the system's response.

As shown in the figure below (Figure 4.1 - Consistency test A), the mean response time of standard lane 0 when traffic consists primarily of trucks is slightly more than 90 seconds, approximately 1.5 times the mean response time of 60 seconds for cars-based traffic, thus confirming our expectations.

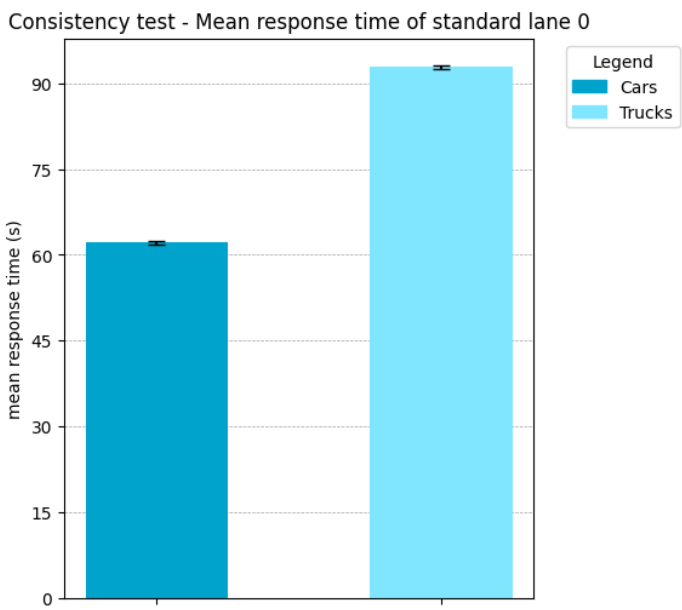


Figure 4.1 - Consistency test A

Consistency test B

This test focuses on the relationship between the arrival rate of vehicles and the service time of lanes. We doubled both the interarrival time of vehicles and the service time of both types of lanes. We expect the same behavior of the system in both cases: in particular, we expect the mean value of cars in queue to be equal in the two tests, given that the ratio between mean service time and mean interarrival time remains constant.

Test B.1

- **T_c** (arrival rate of cars) = 9s,
- **T_t** (arrival rate of trucks) = 90s,
- **S** (mean of the exponential service time distribution of std lanes) = 62s,
- **S_etc** (service time of ETC lanes) = 5s .

Test B.2

- **T_c** (arrival rate of cars) = 18s,
- **T_t** (arrival rate of trucks) = 180s,
- **S** (mean of the exponential service time distribution of std lanes) = 124s,
- **S_etc** (service time of ETC lanes) = 10s .

As shown in the figure below (Figure 4.2 - Consistency test B), both in the standard lane 0 and in the ETC lane 0 the mean number of vehicles is approximately equal in the two tests, thus confirming our expectations.

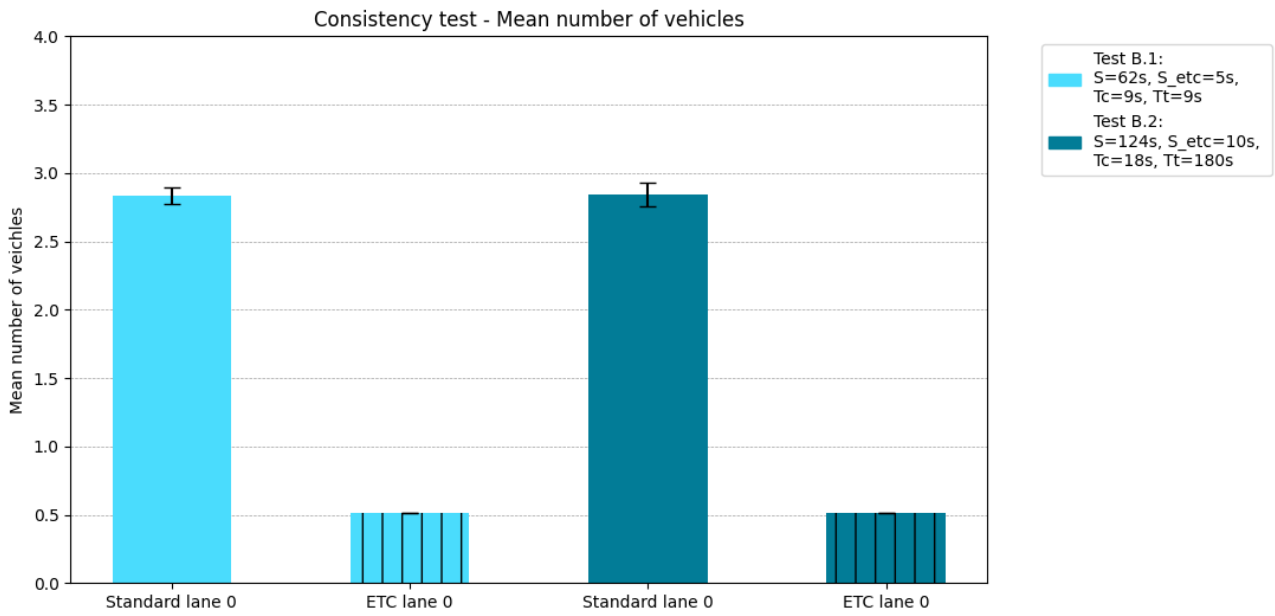


Figure 4.2 - Consistency test B

Continuity test

To evaluate the system's correctness, a test is necessary to verify the sensitivity of the output to small changes in the input parameters. This can be demonstrated by comparing the output response times of four configurations with very similar service times for the standard lanes:

- Test 1: **S** = 53.5s;
- Test 2: **S** = 54s;
- Test 3: **S** = 54.5s;
- Test 4: **S** = 55s.

The other parameters are equal to the ones of the general configuration.

Observing the output in Figure 4.3 - Continuity test, we can see that the mean response time does not change wildly with a slight change of inputs and the chart presents a linear and continuous trend.

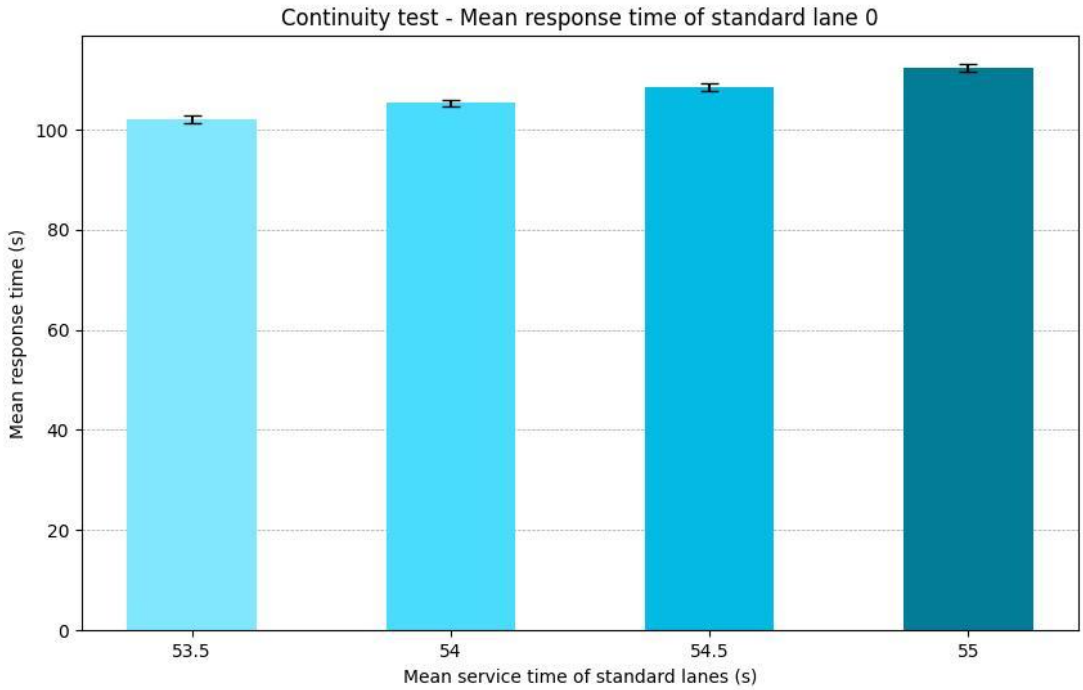


Figure 4.3 - Continuity test

Monotonicity test

Another important experiment to verify the correctness of the system is the monotonicity test, which is useful to assess the monotonicity of some performance indexes using different combinations of factors. In our case we changed the number of total lanes, keeping constant all the other parameters from the general configuration (including K, the number of ETC lanes).

Observing the Figure 4.4 - Monotonicity test, we can conclude that increasing N the mean response time decreases as expected without any strange behaviors.

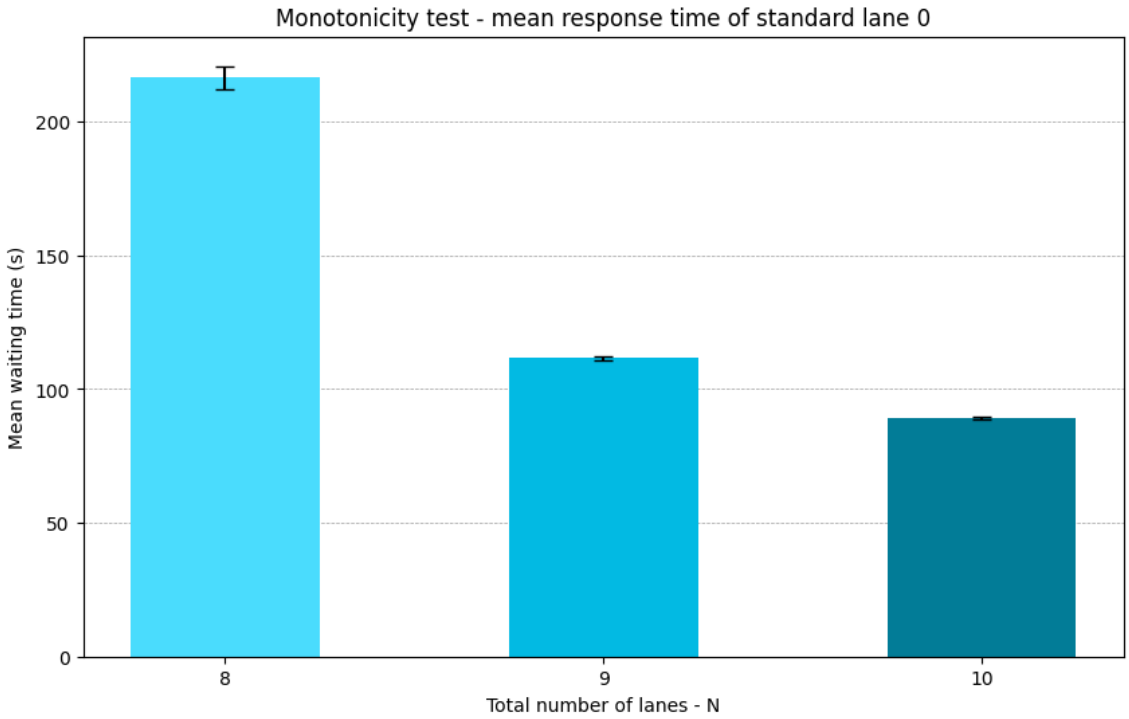


Figure 4.4 - Monotonicity test

5. CALIBRATION

Factors calibration

The aim of this section is to establish the values and intervals of the factors needed to correctly reproduce the behavior of the real system. Most of the factor values were derived from reports provided by *Anas S.p.A*[1], the major Italian company responsible for the maintenance and management of road infrastructure and highways. According to their estimate, the number of vehicles during peak rush hour on a highway is approximately 400 per hour. Additionally, we found that trucks make up about 10% of the total traffic.

Accurate estimates for the p, the percentage of vehicles equipped with the ETC device, are difficult to obtain. We estimated its value by comparing the number of ETC devices sold in Italy to the total number of cars registered, calculating a value of approximately 20% / 30%.

We have decided to fix the total number of lanes N to focus our analysis on the behavior of one specific model, a medium sized highway barrier, by varying the number of ETC lanes K and the workload. In this way we will find the optimal relationship between N and K. This relationship can then be scaled to model larger or smaller highway toll systems.

The only factor not considered in the estimates provided by *Anas* was the mean service time of standard lanes, for which we conducted a small analysis to find the upper limit of the values. In the figure below (Figure 5.1 - Mean service time calibration), it can be observed that the mean response time will increase fast after the mean service time of standard lane exceeds 62s, so we decided to use 62s as the upper limit of our case study.

[1] https://www.stradeanas.it/sites/default/files/Schema_Report_post_2019.pdf

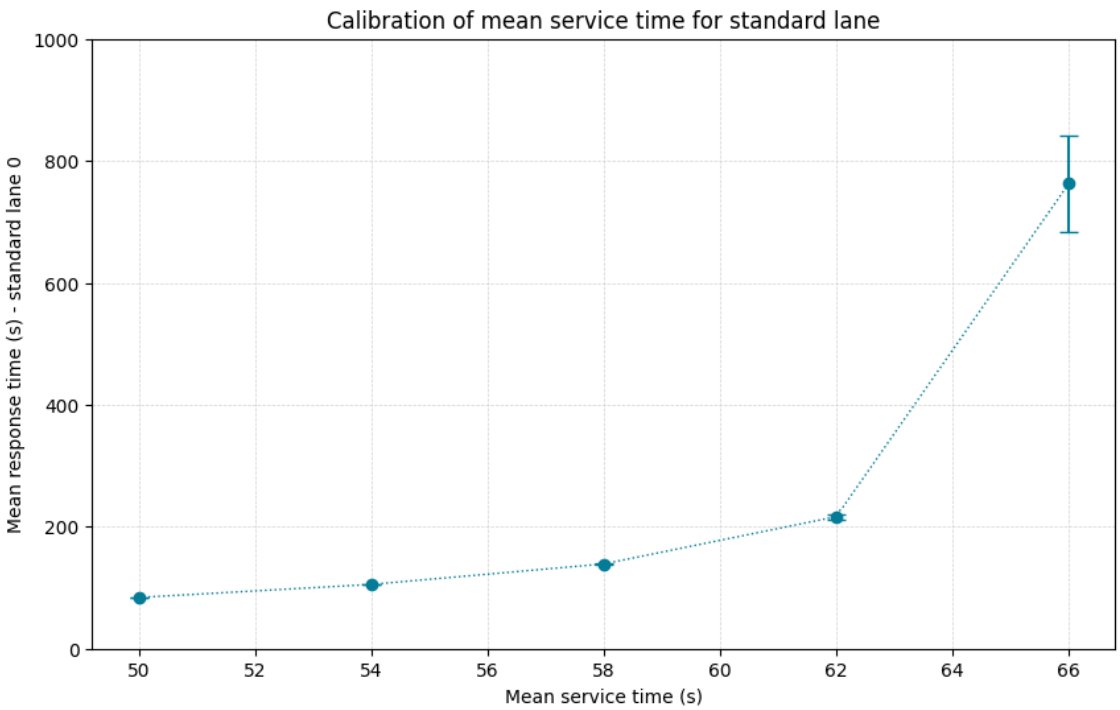


Figure 5.1 - Mean service time calibration

This is the range of values of the factors:

- **S** - Mean service time for standard lanes: [54s, 62s].
- **T_c** - Mean interarrival Time for cars: [9s, 12s].
- **T_t** - Mean interarrival Time for trucks: [90s, 120s].
- **p** - Percentage of vehicles equipped with ETC device: [0.1, 0.6].
- **K** - Number of ETC lanes: [0, 3].
- **S_etc -** Service time for ETC lanes : 5s.
- **stddev** / $\sqrt{\sigma}$ – Standard deviation of the lognormal distribution= [30s, 60s, 90s]. These values correspond to the values of the standard deviation of the underlying (?) Normal distribution.

Calibration of warm-up time and simulation time

In order to calibrate the warm-up time, we studied the evolution of the performance indexes during the early stage of the simulation. We tested various scenarios by changing the values of the factors and determined that a general warm-up time good enough for every scenario under study is 200.000 seconds. In fact, as shown in the graph below (Figure 5.2 - WarmUp time), the behavior of the waiting time before 200.000 seconds is unstable and significantly different from the behavior after the threshold. Although the waiting time continues to oscillate after the threshold, it does so inside a well-defined band.

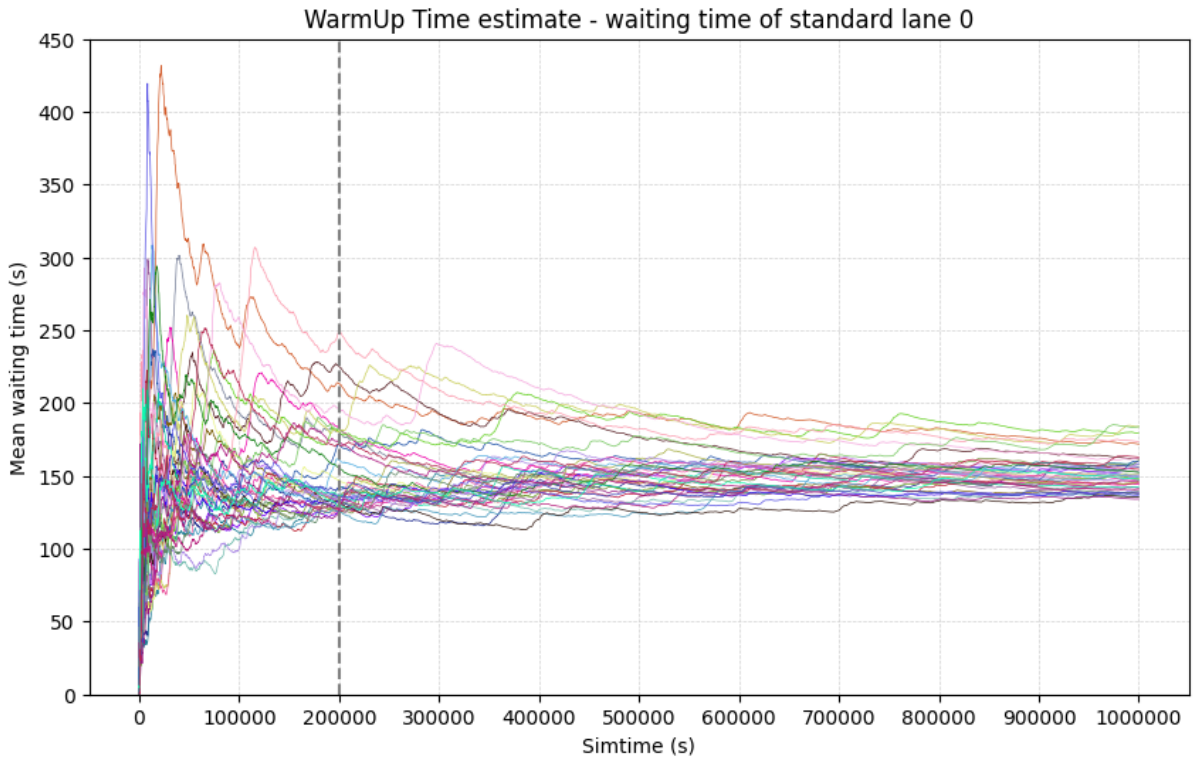


Figure 5.2 - WarmUp time

For all the experiments we conducted, we decided to set the simulation time to 1.000.000 seconds. That would allow us to collect enough data to draw our conclusions.

6. SIMULATION EXPERIMENTS

Notes on experiments

In the following graphs we present performance indexes using data from standard lane 0 and ETC lane 0. As shown in the figures below (Figure 6.1 - Waiting time of standard lanes, Figure 6.2 - Waiting time of ETC lanes), these lanes are always the most occupied, due to the fact that when two or more lanes have the same amount of vehicles queue the arriving vehicle is sent to the lane with the lower index. Using these two lanes for the analysis allows us to focus on the worst-case scenarios in terms of congestion.

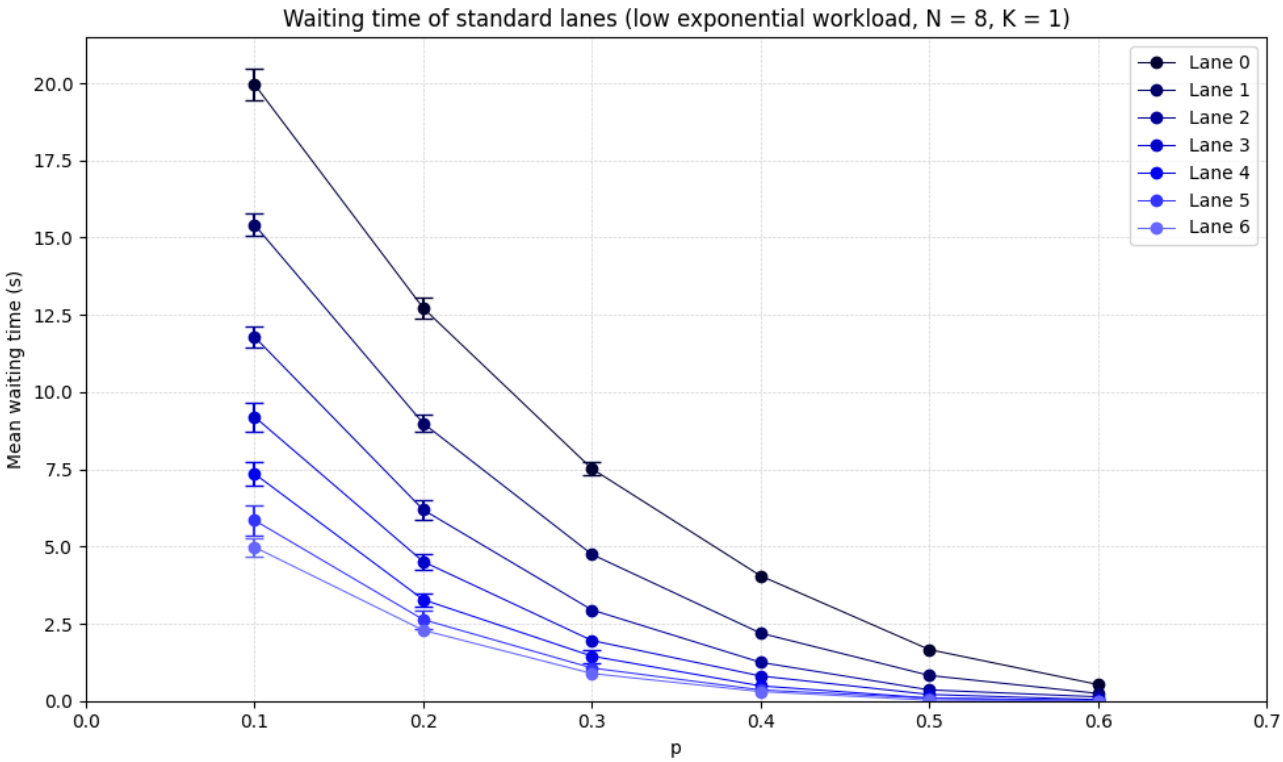


Figure 6.1 - Waiting time of standard lanes

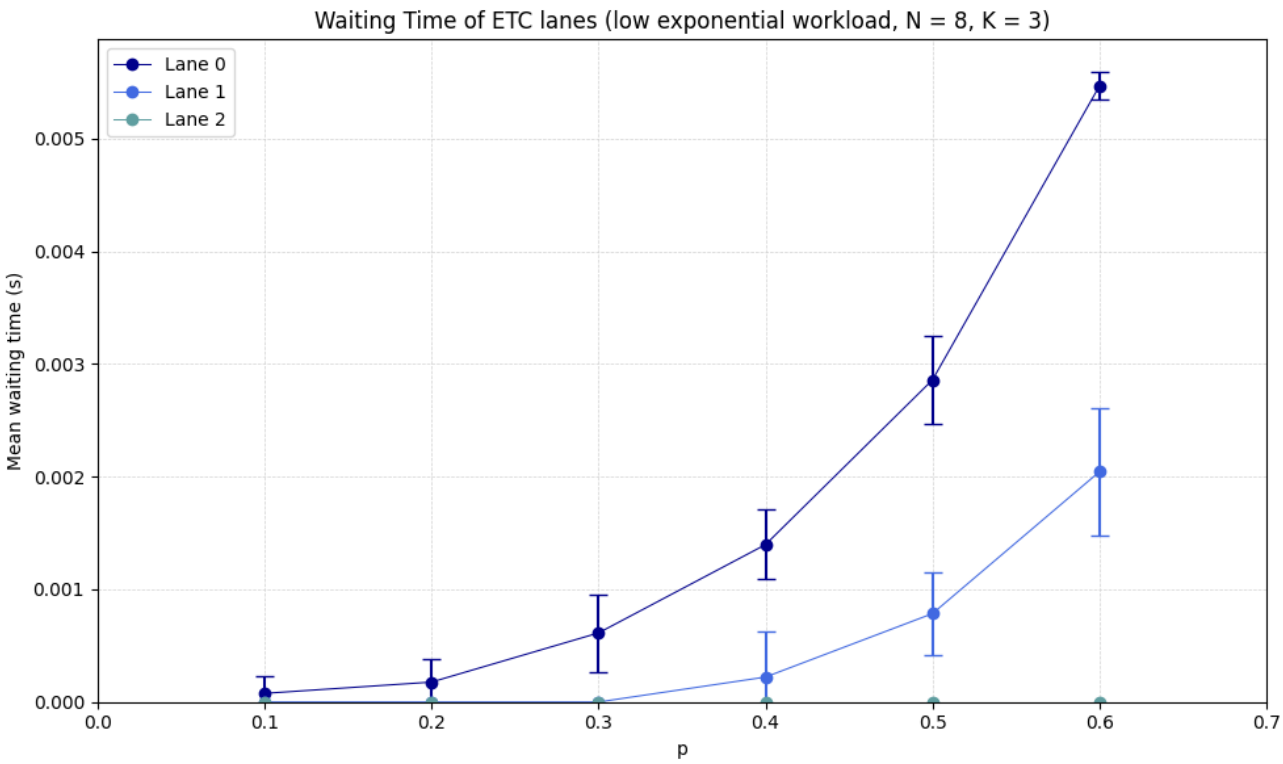


Figure 6.2 - Waiting time of ETC lanes

All single experiments were repeated **35** times.

For all the following graphs we used a 95% confidence interval. We decided not to display the confidence interval when the lines are clustered with their relative marker. In these cases, the absence of a visible confidence interval should be interpreted as its range being smaller than the size of the marker itself, centered around the displayed point.

We simulated all combinations of **p** (the percentage of vehicles equipped with the ETC device) ranging from 0.1 to 0.6 and **K** (number of ETC lanes) ranging from 0 to 3, but only those that are not out of scale are shown in the graphs. Those combinations correspond to experiments where the system becomes unstable: they are irrelevant to our analysis and they would also make the graphs unreadable, thus we decided to omit them.

Exponential distribution

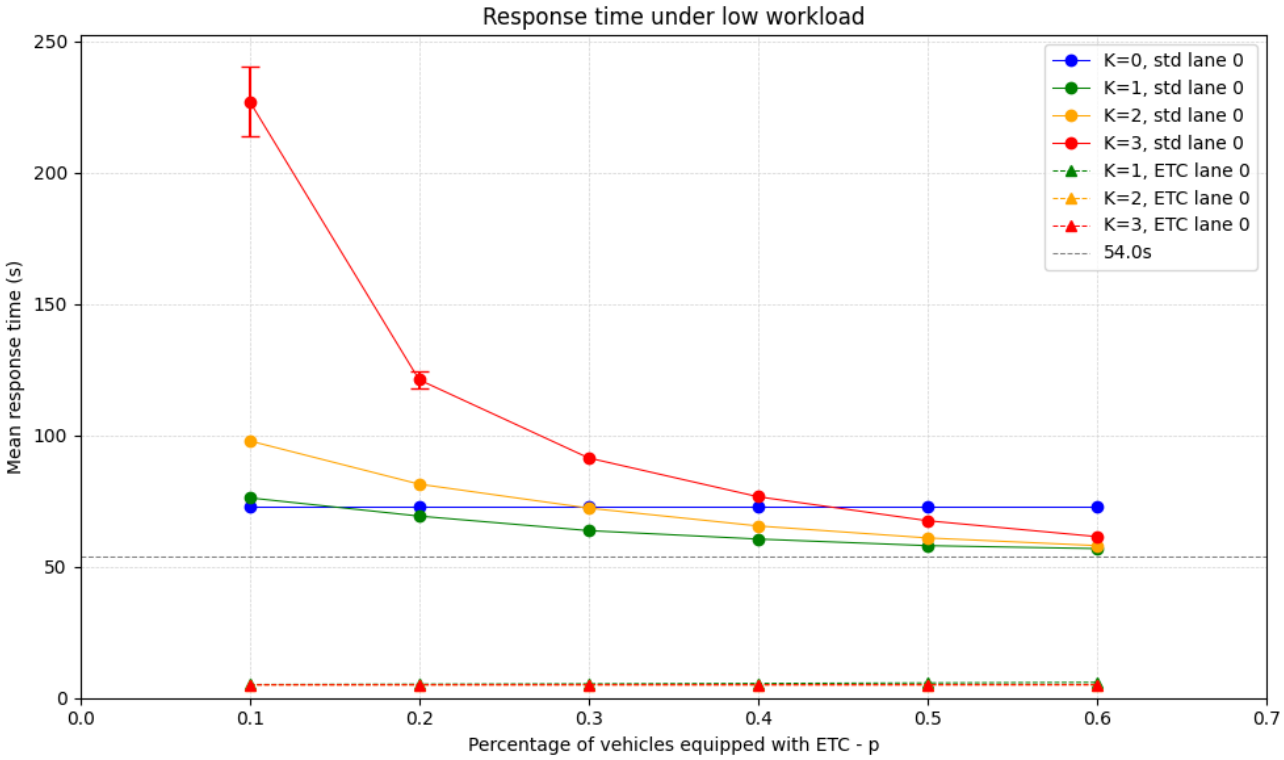
We decided to study the system's behavior under varying workloads. To reduce the number of possible experiments we devised three different workload scenarios: these scenarios combine increasing service times with decreasing interarrival times, creating different levels of congestion. Analyzing all possible combinations of these two factors would be difficult and would also lead to situations with very similar results, as shown in the second consistency test (

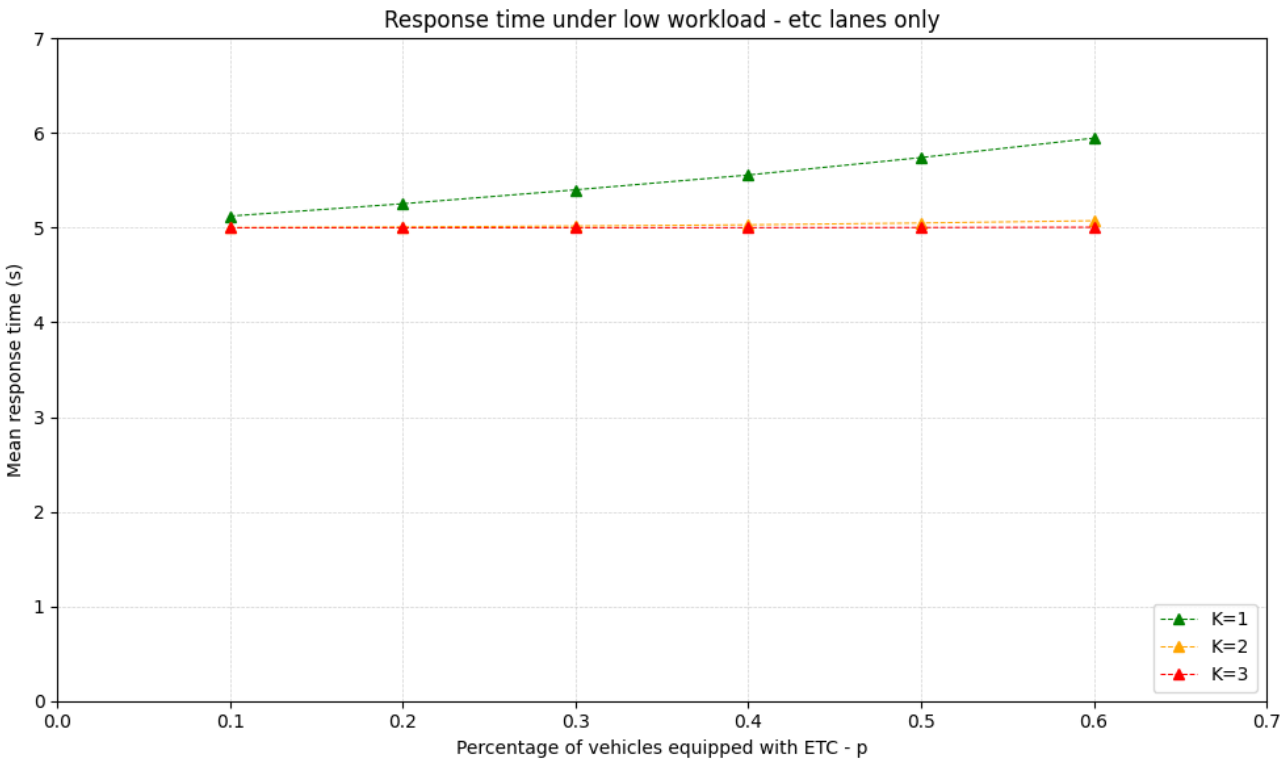
Consistency test B).

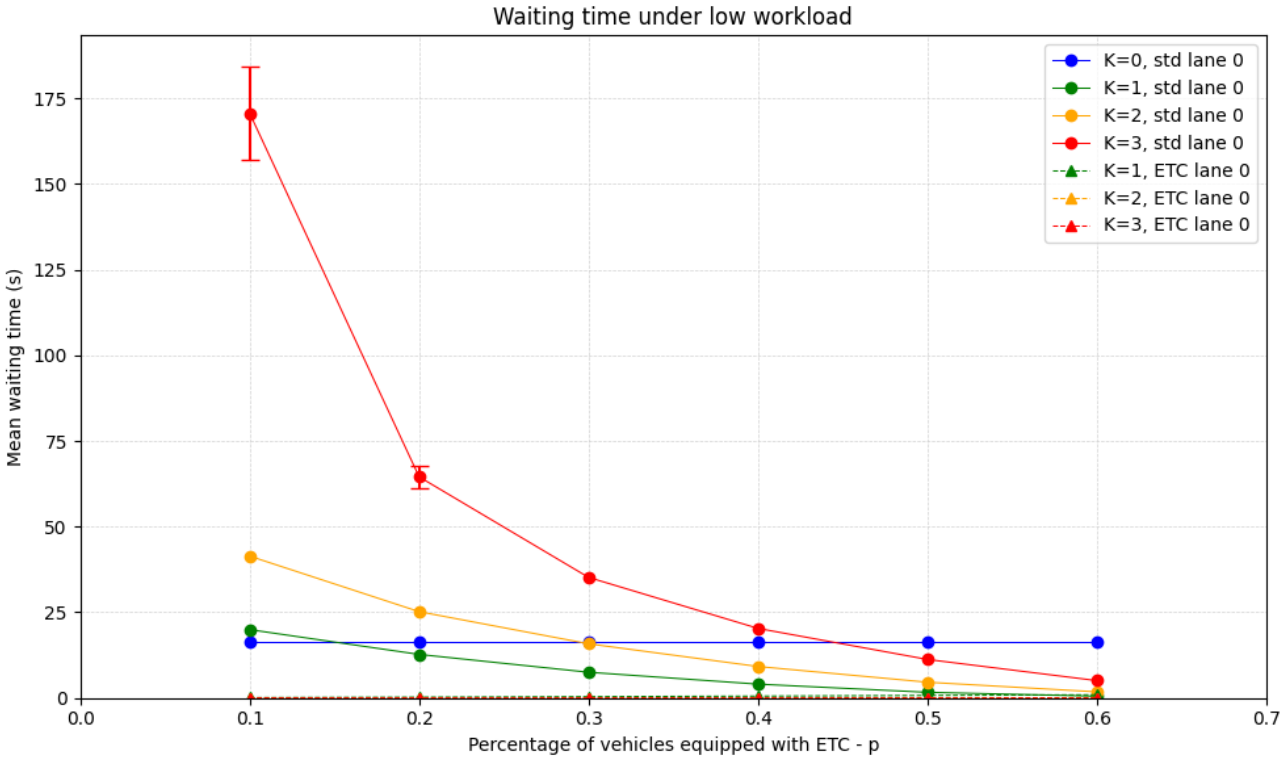
Low workload

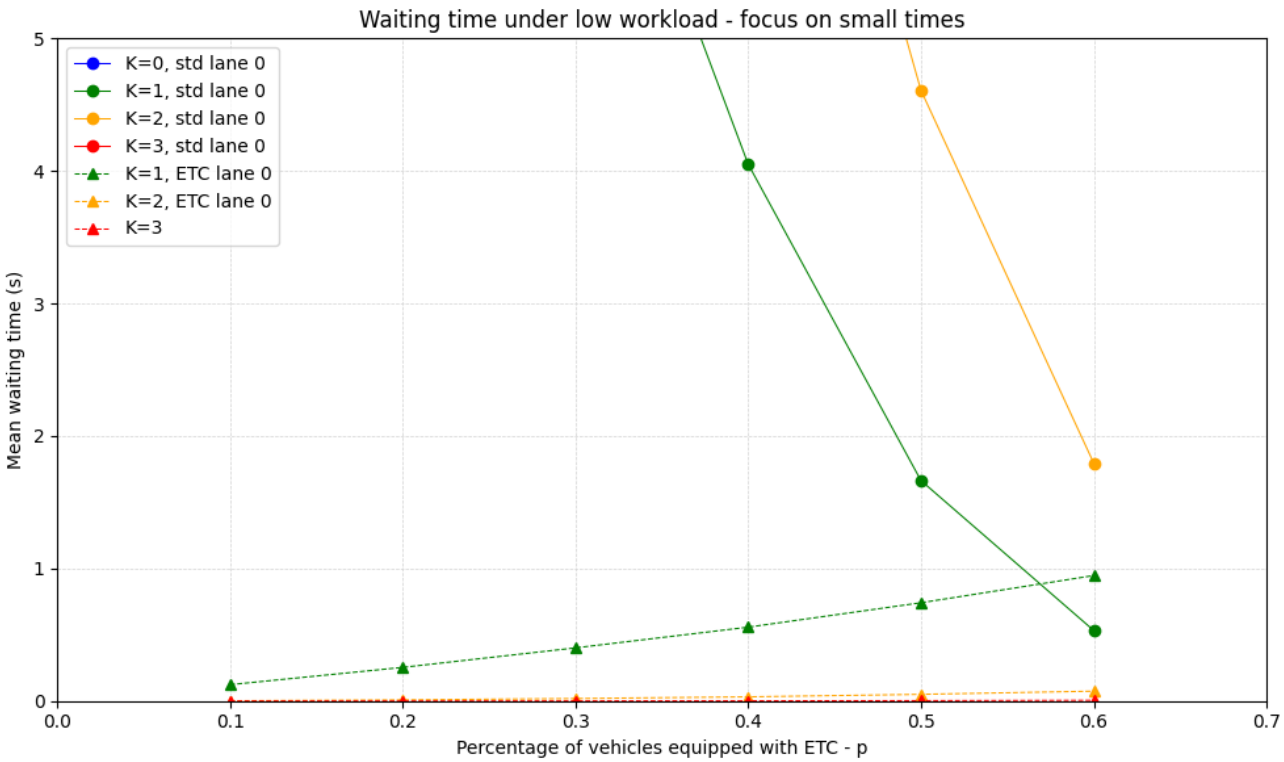
Workload factors:

- $\mathbf{T_c}$ = 12s, an average of 300 cars per hour.
- $\mathbf{T_t}$ = 120s, an average of 30 trucks per hour
- **S** = 54s.


Response time under low workload
Mean response time (s)
Percentage of vehicles equipped with ETC - p
K=0, std lane 0
K=1, std lane 0
K=2, std lane 0
K=3, std lane 0
K=1, ETC lane 0
K=2, ETC lane 0
K=3, ETC lane 0
54.0s


Response time under low workload - etc lanes only
Mean response time (s)
Percentage of vehicles equipped with ETC - p
K=1
K=2
K=3

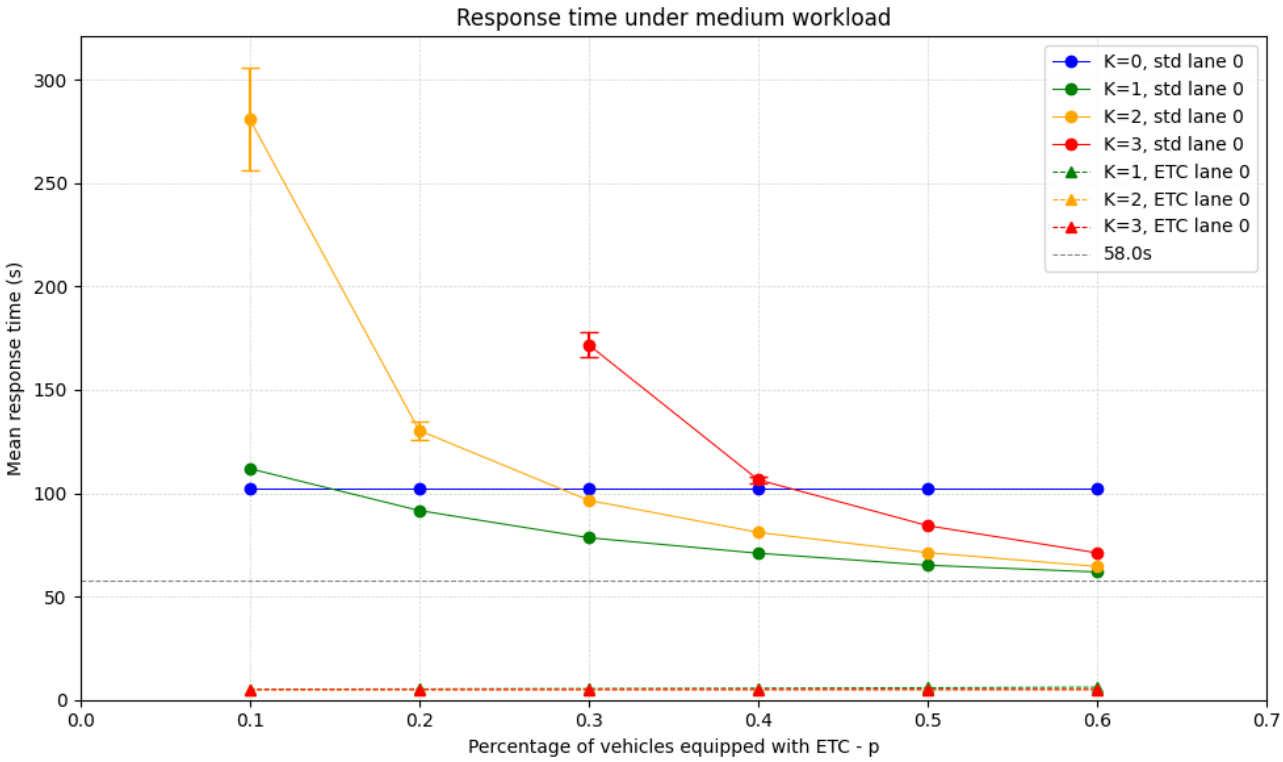

Waiting time under low workload
K=0, std lane 0
K=1, std lane 0
K=2, std lane 0
K=3, std lane 0
K=1, ETC lane 0
K=2, ETC lane 0
K=3, ETC lane 0
Mean waiting time (s)
Percentage of vehicles equipped with ETC - p

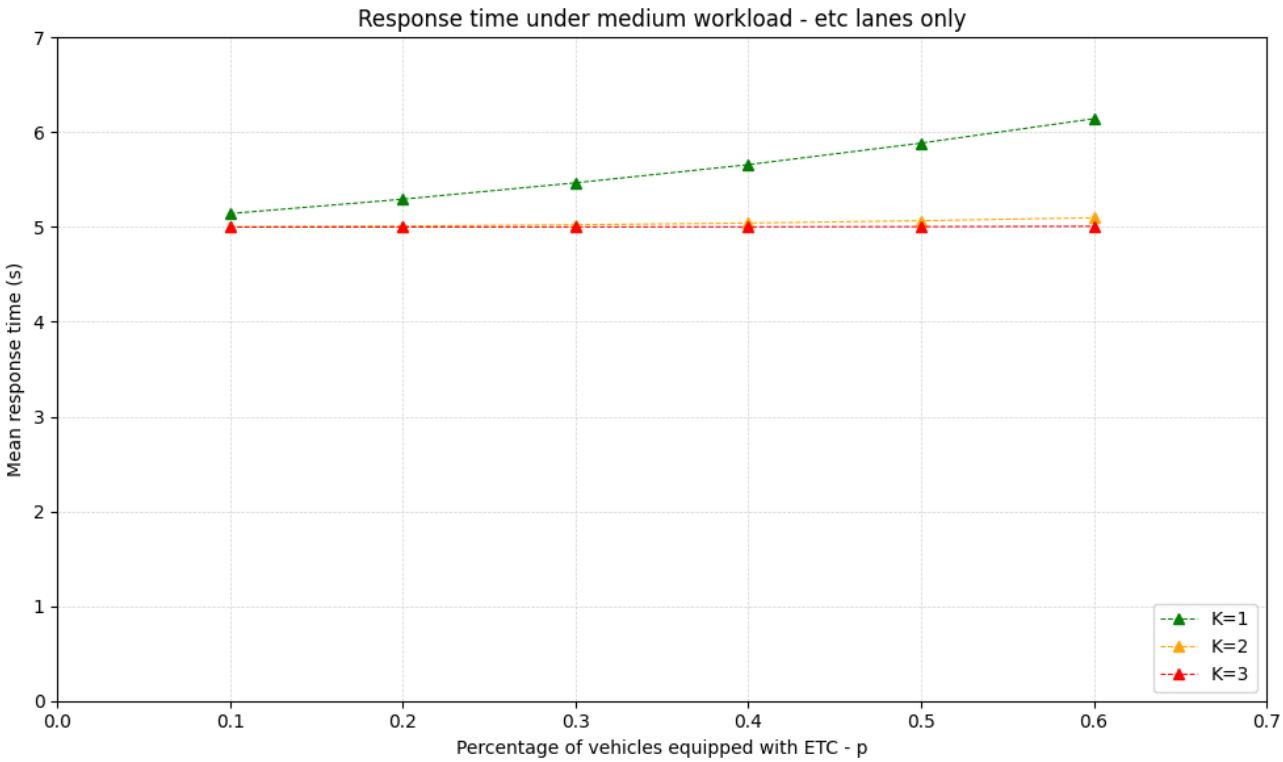

Waiting time under low workload - focus on small times
K=0, std lane 0
K=1, std lane 0
K=2, std lane 0
K=3, std lane 0
K=1, ETC lane 0
K=2, ETC lane 0
K=3
Mean waiting time (s)
Percentage of vehicles equipped with ETC - p

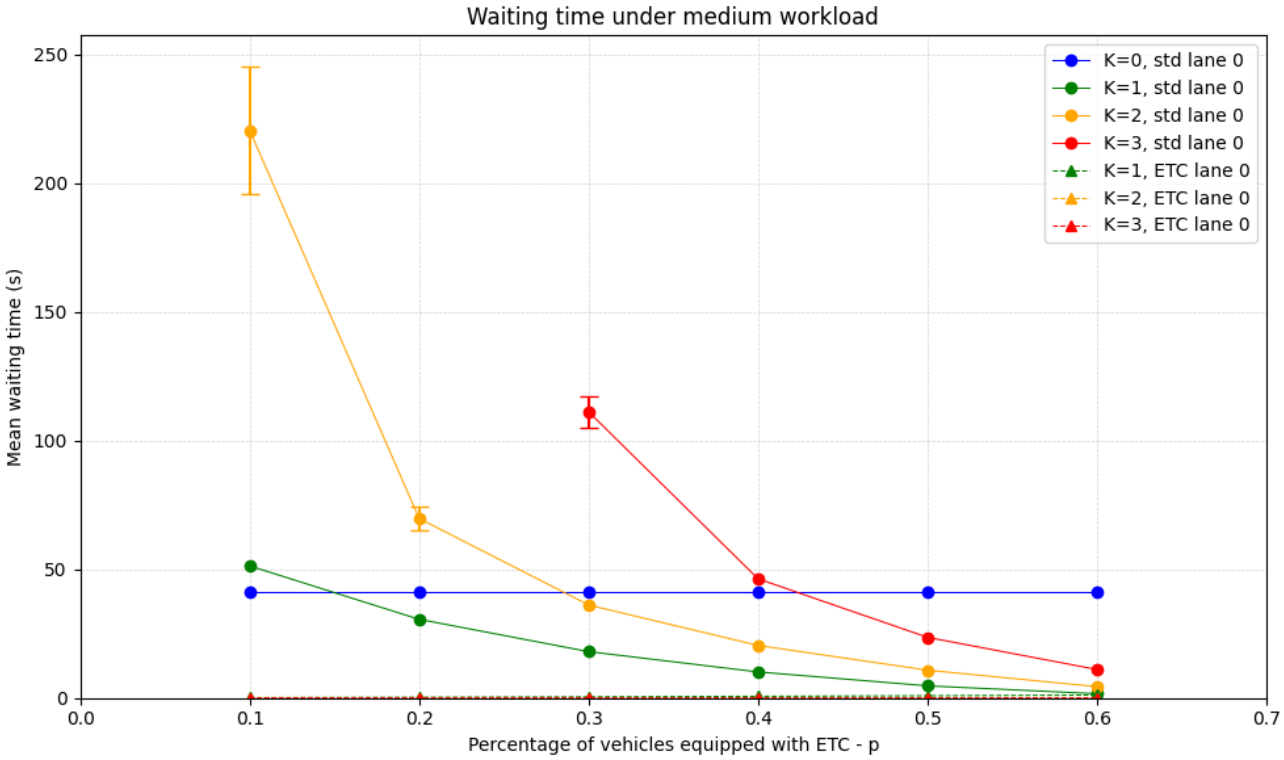
Medium workload

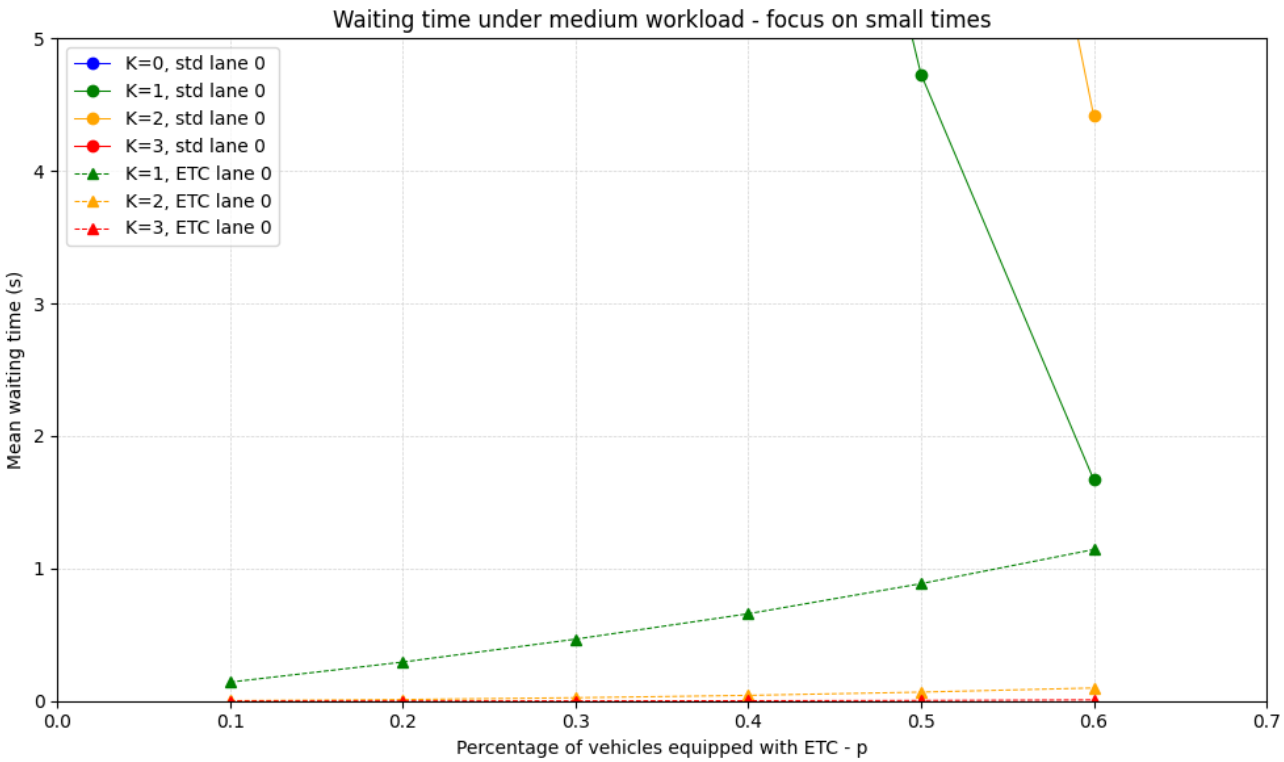
Workload factors:

- **T_c** = 10.5s, an average of 350 cars per hour.
- **T_t** = 105s, an average of 35 trucks per hour.
- **S** = 58s.





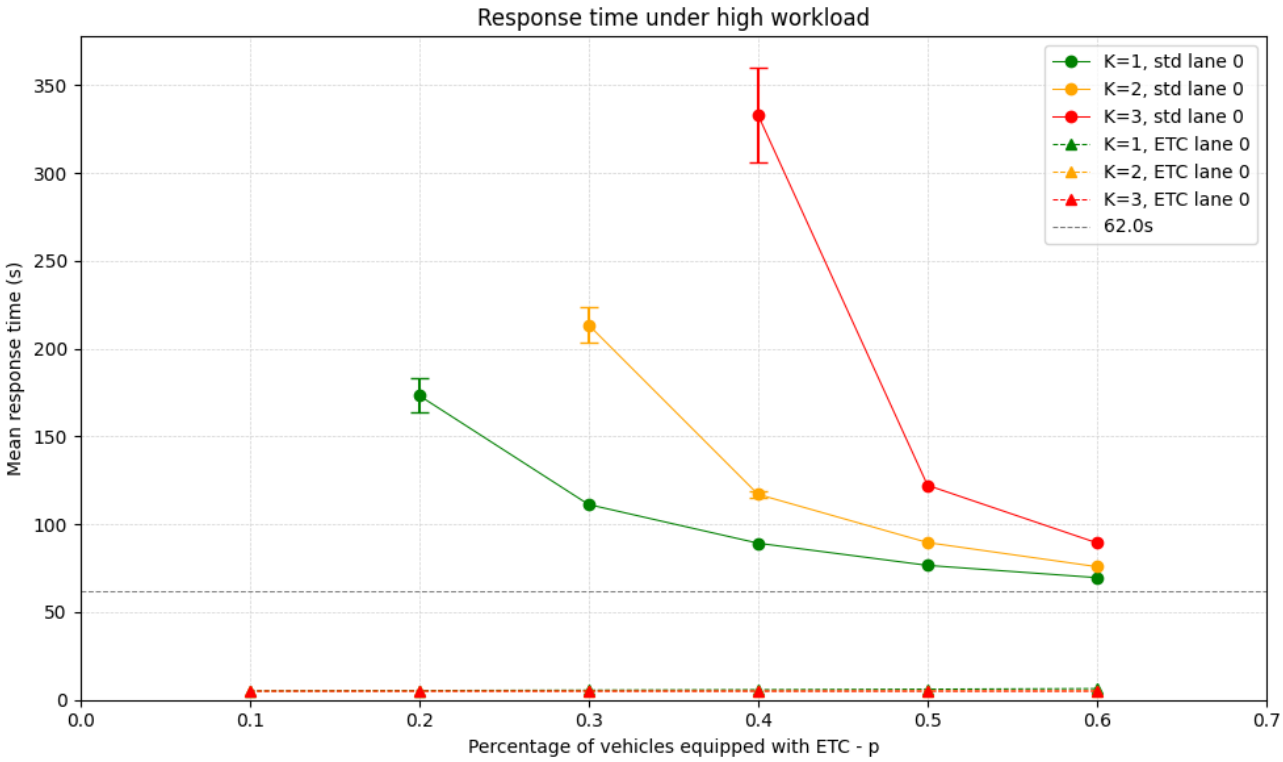

Waiting time under medium workload
Mean waiting time (s)
Percentage of vehicles equipped with ETC - p
K=0, std lane 0
K=1, std lane 0
K=2, std lane 0
K=3, std lane 0
K=1, ETC lane 0
K=2, ETC lane 0
K=3, ETC lane 0
0
50
100
150
200
250
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7

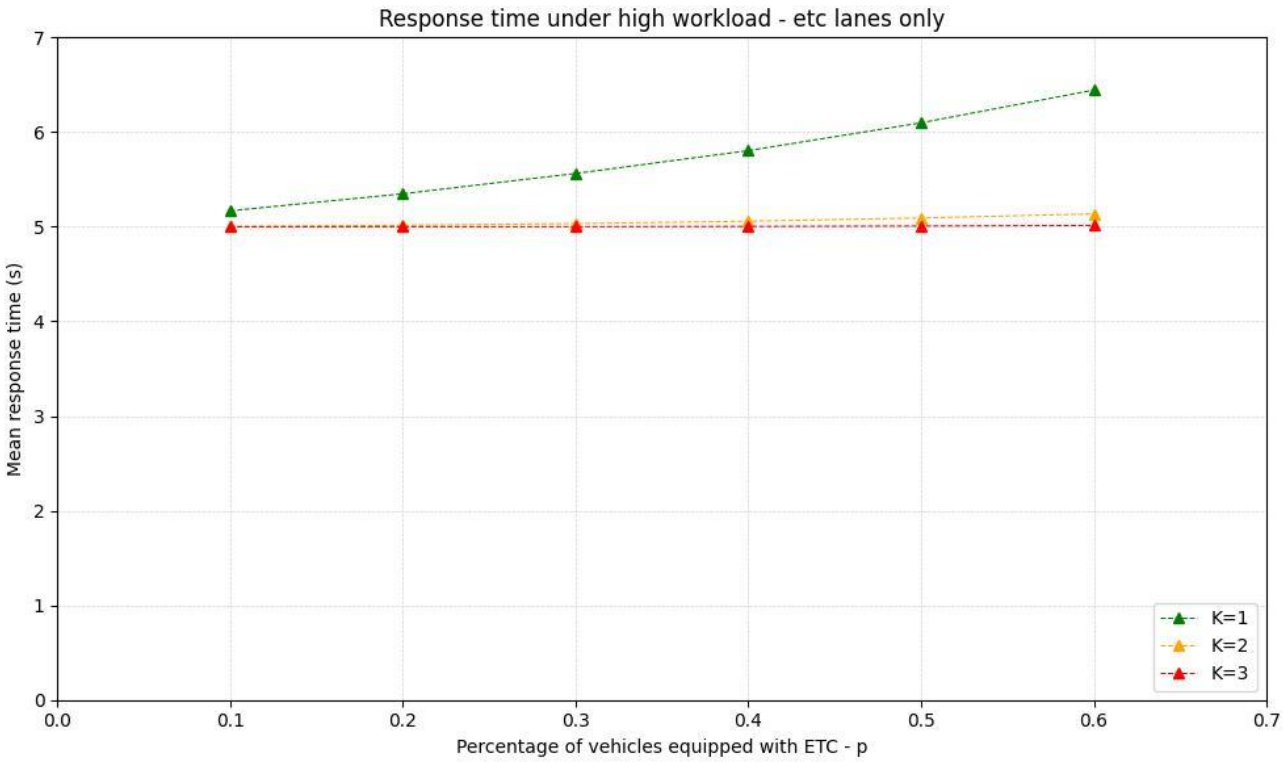

Waiting time under medium workload - focus on small times
Mean waiting time (s)
Percentage of vehicles equipped with ETC - p
K=0, std lane 0
K=1, std lane 0
K=2, std lane 0
K=3, std lane 0
K=1, ETC lane 0
K=2, ETC lane 0
K=3, ETC lane 0
0
1
2
3
4
5
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7

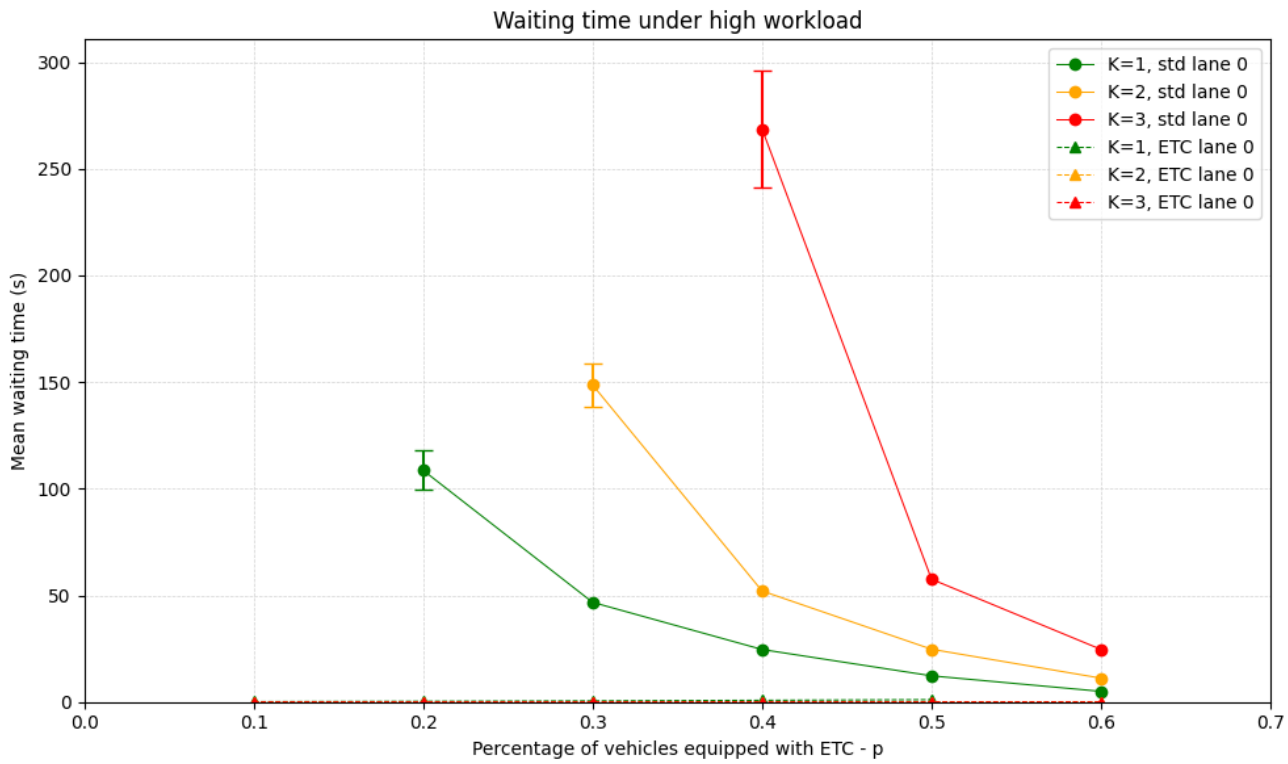
High workload

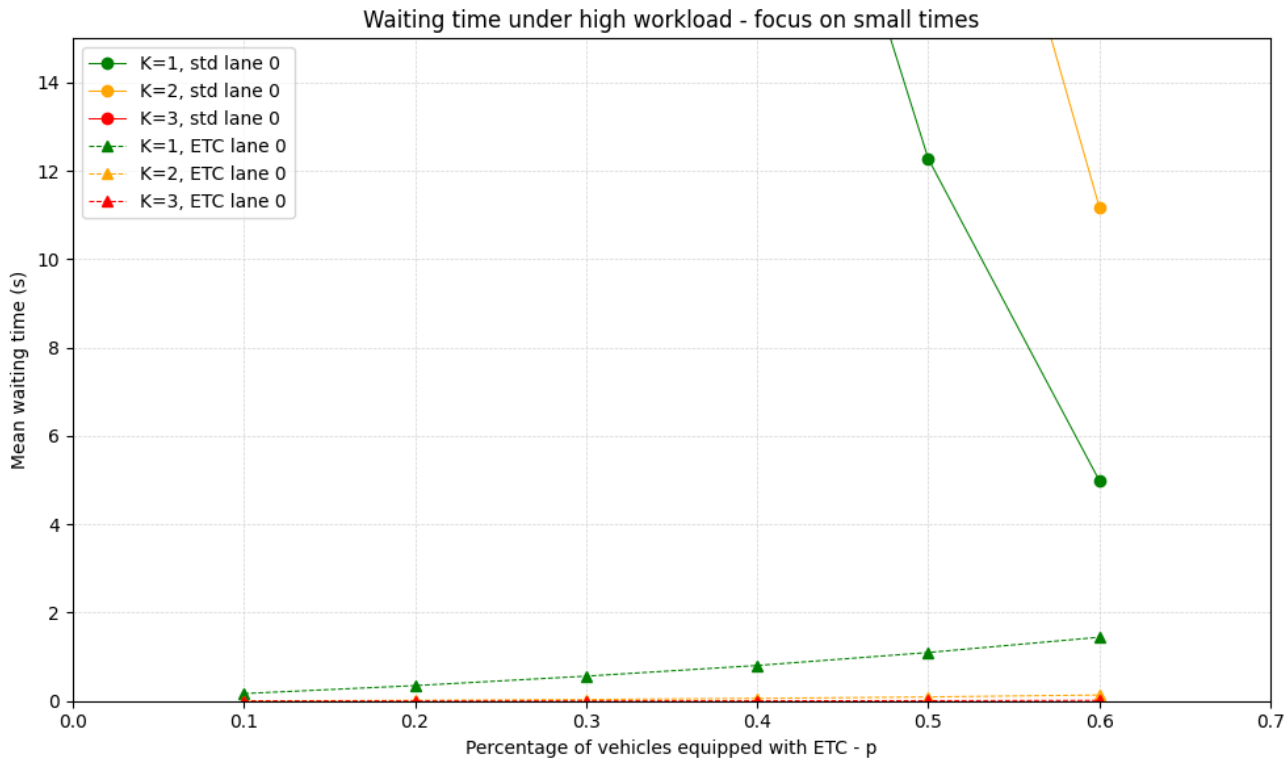
Workload factors:

- **T_c** = 9s, an average of 400 cars per hour.
- **T_t** = 90s, an average of 40 trucks per hour.
- **S** = 62s









In this graphs we decided not to show the standard lane 0 data for the case K = 0 because it exhibits the same behavior as the two previous cases but with significantly higher values. Including it would rescale the entire graph, rendering the other parts difficult to read.

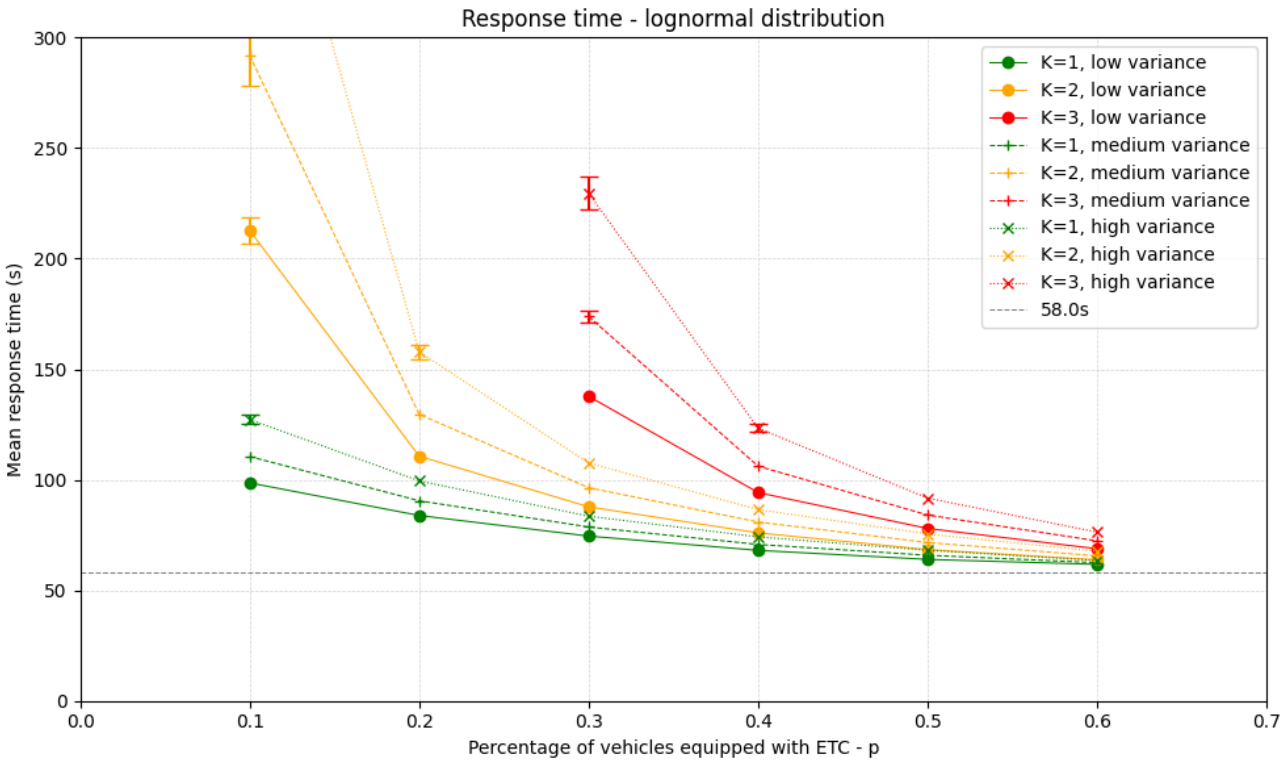
Lognormal distribution

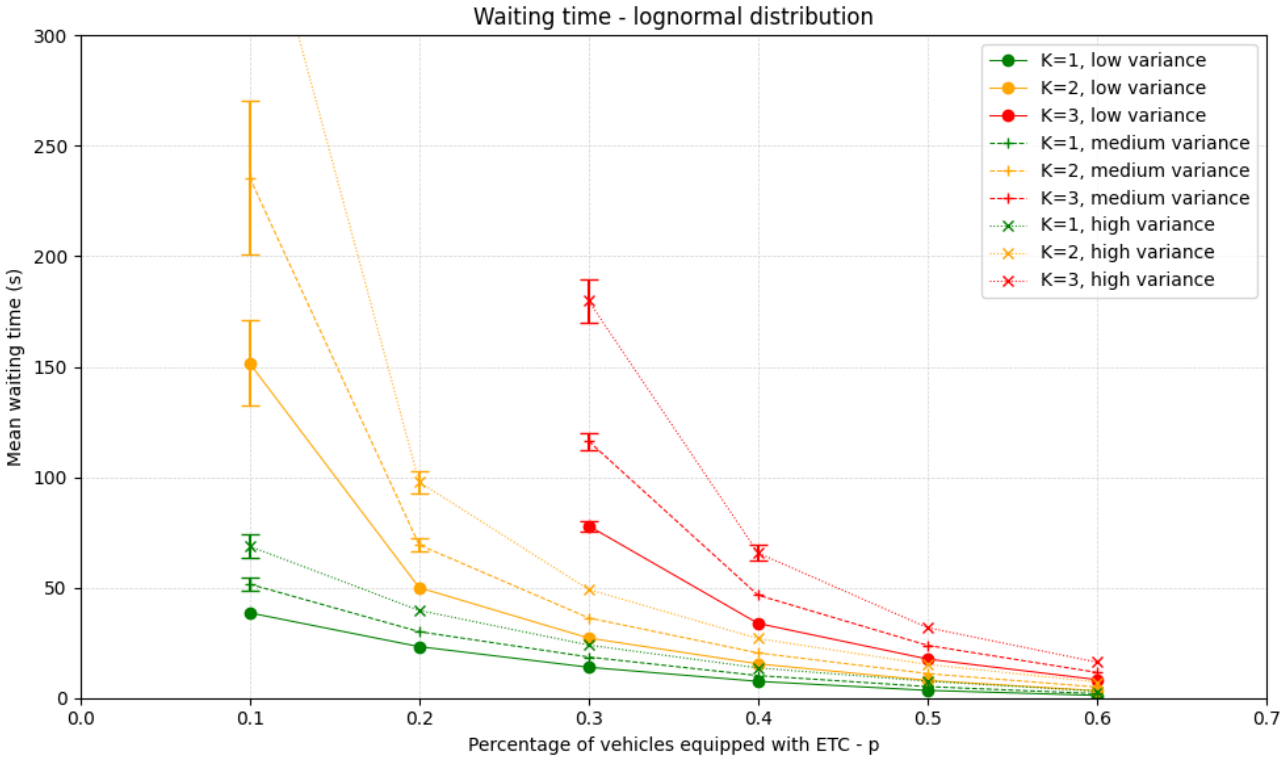
In the graphs related to the experiments based on the lognormal distribution, we decided to plot data only from standard lane 0, as the change in the distribution type affects only the standard lanes.

Workload factors:

- **T_c** = 10.5s, an average of 350 cars per hour.
- **T_t** = 105s, an average of 35 trucks per hour
- **S** = 58s.
- **stddev** = {30s, 60s, 90s}
    - **30s**: Low variance workload
    - **60s**: Medium variance workload
    - **90s**: High variance workload

Moreover, in the experiments based on the lognormal distribution, we excluded the case K = 0 because it exhibits a behavior very similar to that observed in the exponential distribution under medium workload. Furthermore, including it would cluster the graph and reduce clarity.




Waiting time - lognormal distribution
Mean waiting time (s)
Percentage of vehicles equipped with ETC - p
0
50
100
150
200
250
300
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
K=1, low variance
K=2, low variance
K=3, low variance
K=1, medium variance
K=2, medium variance
K=3, medium variance
K=1, high variance
K=2, high variance
K=3, high variance

7. CONCLUSION

As can be seen by comparing the experiments with K = 0 to the other cases, the addition of ETC lanes is fundamental not only for significantly reducing the waiting times for ETC-equipped vehicles but also for improving the overall performance of the system. While adding ETC lanes has a substantial impact on the system, we observed that using high values of K (the number of ETC lanes) leads to inefficient use of lanes. This is because the arrival flow of ETC vehicles depends on p (which is quite low in general) and the service time of ETC lanes is so low that it does not cause significant enough queuing to require the addition of other lanes.

Furthermore, keeping constant the number of total lanes N, the increase of K (number of ETC lanes) could be a risk because N-K standard lanes could be not enough to handle the vehicles incoming, leading to congestion and increasing the waiting times in those standard lanes.

We have a significant improvement even in standard lanes where the percentage p is high, so that a significant part of vehicle flow is assigned to the ETC lane(s).

These observations hold true for every configuration we tested. The change in service time distribution for the standard lane has no noticeable effects, and an increase in workload only increases the demand for standard lanes compared to ETC lanes.

Regarding the lognormal distribution for the standard lane service time we observe that, as expected, an increase in standard deviation results in a general increase in waiting and response times for the standard lanes, without changing the overall behavior of the system under varying K and p.

Comparing our results with real-world scenarios it can be observed that the ratio of one ETC lane over eight total lanes is a reasonable approximation. However there are some aspects not considered in our simulations that could influence this result: traffic in the real world is far from constant, fluctuating both daily, typically featuring two peak periods and very light flow during the night, and seasonally. During peak flow periods the percentage of ETC-equipped vehicles may be higher than average, thus favoring a higher ratio of K to N. Furthermore, the physical layout of the highway barrier could impact congestion, making it more advantageous to have two or more ETC lanes located at opposite ends of the barrier to minimize the need for vehicles to swerve across multiple lanes.